

GED Science Practice Test 1

PART A — QUESTIONS

GED-SCI1-001

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student observes two samples under a microscope. Sample A contains cells with a cell wall, chloroplasts, a nucleus, and a large central vacuole. Sample B contains cells with a nucleus, a cell membrane, and mitochondria, but no cell wall or chloroplasts.

Which conclusion is best supported by these observations?

- A) Sample A contains animal cells, and Sample B contains plant cells.
- B) Sample A contains plant cells, and Sample B contains animal cells.
- C) Both samples contain bacteria because both have cell membranes.
- D) Both samples contain plant cells because both have nuclei.

GED-SCI1-002

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student measures the mass and volume of four objects made of different materials.

Object	Mass (g)	Volume (mL)
A	24	8
B	30	10
C	45	15
D	50	25

Which object has the lowest density?

- A) Object A
- B) Object B
- C) Object C
- D) Object D

GED-SCI1-003

Section: Science

Type: numeric_entry

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A toy car travels 18 meters in 6 seconds at a constant speed.

What is the speed of the toy car in meters per second?

Type your answer as a whole number.

GED-SCI1-004

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A city near the coast often has cooler afternoons than a city farther inland at the same latitude. During the day, land warms more quickly than ocean water. Air above the warmer land rises, and cooler air from over the ocean moves toward the land.

Which process is mainly described in this scenario?

- A) A sea breeze caused by uneven heating
- B) A landslide caused by gravity
- C) A volcanic eruption caused by magma pressure
- D) A solar eclipse caused by the Moon blocking sunlight

GED-SCI1-005

Section: Science

Type: multi_select

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Select TWO answers.

A person's body temperature rises after an infection. The immune system responds by increasing the activity of white blood cells. The body also sweats when the temperature becomes too high.

Which TWO statements describe homeostasis in this scenario?

- A) Sweating helps cool the body when temperature increases.
- B) White blood cells help the body respond to infection.
- C) The body stops using energy during an infection.
- D) A fever always means the body has lost all regulation.
- E) The immune response prevents all future infections.

GED-SCI1-006

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student wants to test how the temperature of water affects how quickly sugar dissolves. The student uses 200 mL of water, 10 g of sugar, and the same stirring speed in each trial. Only the water temperature changes.

What is the independent variable in this experiment?

- A) The amount of sugar
- B) The volume of water
- C) The temperature of the water
- D) The stirring speed

GED-SCI1-007

Section: Science

Type: dropdown

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

In a simple food chain, grass is eaten by rabbits, and rabbits are eaten by foxes. Grass gets energy from sunlight. In this food chain, grass is the [D1], and foxes are [D2].

Dropdowns:

D1) producer | decomposer | predator

D2) primary consumers | secondary consumers | producers

GED-SCI1-008

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A sealed container holds vinegar and baking soda before they react. The total mass of the container and chemicals is 156 g. After the reaction, bubbles form, but the container stays sealed.

What should the total mass be after the reaction is complete?

- A) Less than 156 g because gas has no mass
- B) Exactly 156 g because the container is sealed
- C) More than 156 g because bubbles add new matter
- D) Impossible to predict because all reactions change mass

GED-SCI1-009

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A gardener records the average height of bean plants grown under different amounts of daily light.

Daily light (hours): 2, 4, 6, 8

Average plant height (cm): 7, 13, 19, 24

Which conclusion is best supported by the data?

- A) More daily light was associated with taller bean plants.
- B) Bean plants grew best with 2 hours of daily light.
- C) Daily light had no effect on average plant height.
- D) Bean plants stopped growing after 4 hours of light.

GED-SCI1-010

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A weather station records rainfall for four months.

Month	Rainfall (cm)
March	8
April	12
May	10
June	6

What was the average monthly rainfall for these four months?

- A) 8 cm
- B) 9 cm
- C) 10 cm
- D) 12 cm

GED-SCI1-011

Section: Science

Type: drag_drop

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Match each target with the correct tile.

Tiles:

- A) Use the same type of ramp in every trial
- B) Change the height of the ramp
- C) Measure how far the ball rolls after leaving the ramp
- D) Use a different ball material each time

Targets:

- T1) Independent variable
- T2) Dependent variable
- T3) Controlled variable

GED-SCI1-012

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

In a certain species of flower, purple petals are dominant over white petals. A plant with two dominant alleles is crossed with a plant with two recessive alleles.

What will be the expected petal color of all offspring?

- A) Purple, because each offspring receives at least one dominant allele
- B) White, because each offspring receives one recessive allele
- C) Half purple and half white, because one parent is white
- D) A blend of purple and white, because the alleles mix equally

GED-SCI1-013

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A metal spoon feels colder than a wooden spoon at the same room temperature. Metal transfers thermal energy away from a person's hand faster than wood does.

Which statement best explains why the metal spoon feels colder?

- A) The metal spoon has a lower actual temperature than the wooden spoon.
- B) The metal spoon absorbs cold from the air.
- C) The metal spoon conducts heat away from the hand more quickly.
- D) The wooden spoon creates heat when touched.

GED-SCI1-014

Section: Science

Type: numeric_entry

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

In a pond survey, 80 organisms are counted. Of these organisms, 20 are snails.

What percent of the counted organisms are snails?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI1-015

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

At a plate boundary, one oceanic plate moves beneath another plate. This process can produce deep ocean trenches, earthquakes, and volcanic activity.

Which term describes this process?

- A) Deposition
- B) Subduction
- C) Condensation
- D) Weathering

GED-SCI1-016

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student tests whether salt affects the freezing point of water. Four cups each contain 100 mL of water. The student adds different amounts of salt to each cup, places all cups in the same freezer, and records the temperature at which each cup freezes.

Which setup best improves the reliability of the experiment?

- A) Use a different freezer for each cup
- B) Repeat each salt amount in multiple trials
- C) Change both the salt amount and water volume
- D) Record only the cup that freezes first

GED-SCI1-017

Section: Science

Type: dropdown

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

When a flashlight is turned on, chemical energy stored in the batteries is converted mostly into [D1] energy and then into [D2] energy from the bulb.

Dropdowns:

D1) electrical | nuclear | gravitational

D2) light | sound | elastic

GED-SCI1-018

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Researchers study a new medicine designed to reduce blood pressure. One group of patients receives the medicine, while a second similar group receives a pill with no active ingredient. Neither group knows which pill they received.

Why is the second group important?

- A) It allows researchers to compare results against a control group.
- B) It guarantees that the medicine will work for every patient.
- C) It removes the need to measure blood pressure.
- D) It makes the medicine stronger in the first group.

GED-SCI1-019

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student pulls a cart with different forces and records the cart's acceleration.

Force (N): 5, 10, 15, 20

Acceleration (m/s²): 1, 2, 3, 4

Based on the table, what happens to acceleration when force doubles from 10 N to 20 N?

- A) It decreases by half.
- B) It stays the same.
- C) It doubles.
- D) It increases by 10 times.

GED-SCI1-020

Section: Science

Type: multi_select

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

Select TWO answers.

A community tracks changes near a river after trees are removed from a hillside.

Observation | Before tree removal | After tree removal

Average soil loss per storm | Low | High

River water clarity | Clear | Cloudy

Number of small fish observed | 42 | 23

Which TWO conclusions are best supported by the data?

- A) Tree removal was associated with increased soil erosion.
- B) Cloudier river water may have affected fish habitat.
- C) Removing trees caused the river to stop flowing.
- D) Fish numbers increased after tree removal.
- E) Soil loss stayed the same after tree removal.

GED-SCI1-021

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A population of insects includes individuals with green, brown, and yellow body colors. Birds more easily see yellow insects on tree bark. After many generations, the population contains fewer yellow insects and more brown insects.

Which explanation best describes this change?

- A) Yellow insects changed color because they wanted to survive.
- B) Brown insects were more likely to survive and reproduce in that environment.
- C) Birds caused every insect to develop the same mutation.
- D) The insects stopped passing traits to their offspring.

GED-SCI1-022

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student records how far a sound wave travels in air.

Time (s): 1, 2, 3, 4

Distance (m): 343, 686, 1029, 1372

Which statement best describes the relationship in the data?

- A) Distance decreases as time increases.
- B) Distance remains constant as time increases.
- C) Distance is directly proportional to time.
- D) Distance increases randomly with time.

GED-SCI1-023

Section: Science

Type: numeric_entry

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A rain barrel contains 12 liters of water on Monday. By Friday, it contains 18 liters of water.

What is the percent increase in the amount of water?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI1-024

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A doctor compares resting heart rates for four patients.

Patient | Resting heart rate (beats/min)

A | 62

B | 74

C | 81

D | 69

Which patient has the highest resting heart rate?

A) Patient A

B) Patient B

C) Patient C

D) Patient D

GED-SCI1-025

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

Two students push the same box across a floor. The box moves more easily after wheels are attached to its bottom. The wheels reduce the force that opposes motion between the box and the floor.

Which force is mainly reduced by adding wheels?

A) Gravity

B) Magnetism

C) Friction

D) Buoyancy

GED-SCI1-026

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A scientist studies how fertilizer affects algae growth in pond water. Four tanks receive different fertilizer amounts. All tanks receive the same amount of light, the same starting volume of pond water, and the same starting number of algae cells.

Which factor is being controlled?

- A) Amount of fertilizer
- B) Amount of light
- C) Final number of algae cells
- D) Growth rate of algae

GED-SCI1-027

Section: Science

Type: dropdown

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

Water evaporates from a lake, forms clouds, and later falls as rain. In this process, evaporation changes liquid water into [D1], and precipitation returns water to Earth as [D2].

Dropdowns:

D1) water vapor | ice crystals | liquid runoff

D2) rain or snow | underground rock | solar energy

GED-SCI1-028

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A circuit contains a battery, wires, and a bulb. A student adds more bulbs in series, one at a time. The brightness of each bulb decreases as more bulbs are added.

Which explanation best fits the observation?

- A) Adding bulbs in series increases total resistance in the circuit.
- B) Adding bulbs in series removes the battery from the circuit.
- C) More bulbs create extra energy in the wires.
- D) The bulbs stop needing electric current.

GED-SCI1-029

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A class records the number of bacteria colonies growing on plates kept at different temperatures.

Temperature (°C): 5, 20, 35, 50

Colonies after 24 hours: 2, 18, 46, 5

Which temperature produced the greatest bacterial growth?

- A) 5°C
- B) 20°C

- C) 35°C
- D) 50°C

GED-SCI1-030

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student claims that increasing the surface area of a solid makes it react faster with an acid. The student compares one large cube of chalk with the same mass of crushed chalk in equal amounts of acid.

Which observation would best support the claim?

- A) The crushed chalk reacts completely in less time than the large cube.
- B) The large cube and crushed chalk react at exactly the same rate.
- C) The large cube produces no bubbles because it contains no matter.
- D) The crushed chalk gains mass before the reaction starts.

GED-SCI1-031

Section: Science

Type: multi_select

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student investigates whether exercise affects breathing rate. The student measures breathing rate before exercise and immediately after one minute of jumping jacks.

Which TWO changes would make the investigation stronger?

- A) Test several students instead of only one student.
- B) Use the same exercise time for every trial.
- C) Change the counting method for each student.
- D) Record only the highest breathing rate.
- E) Let each student choose a different exercise.

GED-SCI1-032

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares the average distance of four planets from the Sun.

Planet | Average distance from Sun (million km)

Mercury | 58

Venus | 108

Earth | 150

Mars | 228

Which planet listed is farthest from the Sun?

- A) Mercury
- B) Venus
- C) Earth
- D) Mars

GED-SCI1-033

Section: Science

Type: drag_drop

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Match each target with the correct tile.

Tiles:

- A) Breaks down food and absorbs nutrients
- B) Moves oxygen and nutrients through the blood
- C) Exchanges O₂ and CO₂ with the environment
- D) Sends electrical signals to coordinate responses

Targets:

- T1) Respiratory system
- T2) Circulatory system
- T3) Digestive system

GED-SCI1-034

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student heats 100 g of water and records the temperature.

Time heated (min): 0, 2, 4, 6

Temperature (°C): 21, 31, 41, 51

What is the average rate of temperature increase from 0 to 6 minutes?

- A) 3°C/min
- B) 5°C/min
- C) 10°C/min
- D) 30°C/min

PART B — ANSWER KEY + EXPLANATIONS

GED-SCI1-001 — Correct: B — Correct Answer: Sample A contains plant cells, and Sample B contains animal cells. — Explanation: Cell walls, chloroplasts, and large central vacuoles are typical plant-cell structures, while animal cells lack cell walls and chloroplasts.

GED-SCI1-002 — Correct: D — Correct Answer: Object D — Explanation: Density = mass ÷ volume. Object D has density $50 \div 25 = 2$ g/mL, which is lower than the 3 g/mL density of the other objects.

GED-SCI1-003 — Correct: 3 — Tolerance: exact — Correct Answer: 3 — Explanation: $\text{Speed} = \text{distance} \div \text{time} = 18 \text{ m} \div 6 \text{ s} = 3 \text{ m/s}$.

GED-SCI1-004 — Correct: A — Correct Answer: A sea breeze caused by uneven heating — Explanation: Land warms faster than ocean water, causing air movement from the cooler ocean toward the warmer land.

GED-SCI1-005 — Correct: A,B — Correct Answer: Sweating helps cool the body when temperature increases. | White blood cells help the body respond to infection. — Explanation: Sweating helps regulate temperature, and white blood cells help maintain internal stability by responding to infection.

GED-SCI1-006 — Correct: C — Correct Answer: The temperature of the water — Explanation: The independent variable is the factor intentionally changed by the student.

GED-SCI1-007 — Correct: D1=producer; D2=secondary consumers — Correct Answer: D1=producer; D2=secondary consumers — Explanation: Grass makes its own food, so it is a producer. Foxes eat rabbits, so foxes are secondary consumers.

GED-SCI1-008 — Correct: B — Correct Answer: Exactly 156 g because the container is sealed — Explanation: In a sealed container, matter cannot escape, so total mass is conserved.

GED-SCI1-009 — Correct: A — Correct Answer: More daily light was associated with taller bean plants. — Explanation: The table shows plant height increasing as daily light increases.

GED-SCI1-010 — Correct: B — Correct Answer: 9 cm — Explanation: The average rainfall is $(8 + 12 + 10 + 6) \div 4 = 36 \div 4 = 9 \text{ cm}$.

GED-SCI1-011 — Correct: T1=B; T2=C; T3=A — Correct Answer: T1=Change the height of the ramp; T2=Measure how far the ball rolls after leaving the ramp; T3=Use the same type of ramp in every trial — Explanation: The independent variable is changed, the dependent variable is measured, and controlled variables are kept the same.

GED-SCI1-012 — Correct: A — Correct Answer: Purple, because each offspring receives at least one dominant allele — Explanation: A plant with two dominant alleles crossed with a plant with two recessive alleles produces offspring with one dominant allele, so all show the dominant purple trait.

GED-SCI1-013 — Correct: C — Correct Answer: The metal spoon conducts heat away from the hand more quickly. — Explanation: Metal feels colder because it transfers thermal energy away from skin faster than wood does.

GED-SCI1-014 — Correct: 25 — Tolerance: exact — Correct Answer: 25 — Explanation: $20 \div 80 = 0.25$, which is 25%.

GED-SCI1-015 — Correct: B — Correct Answer: Subduction — Explanation: Subduction occurs when one tectonic plate moves beneath another plate.

GED-SCI1-016 — Correct: B — Correct Answer: Repeat each salt amount in multiple trials — Explanation: Repeating trials makes results more reliable by reducing the effect of random error.

GED-SCI1-017 — Correct: D1=electrical; D2=light — Correct Answer: D1=electrical; D2=light — Explanation: Batteries provide electrical energy, which the bulb converts mostly into light and some heat.

GED-SCI1-018 — Correct: A — Correct Answer: It allows researchers to compare results against a control group. — Explanation: A placebo group serves as a control group, helping researchers determine whether the medicine caused the observed effect.

GED-SCI1-019 — Correct: C — Correct Answer: It doubles. — Explanation: Acceleration increases from 2 m/s^2 to 4 m/s^2 when force doubles from 10 N to 20 N.

GED-SCI1-020 — Correct: A,B — Correct Answer: Tree removal was associated with increased soil erosion. | Cloudier river water may have affected fish habitat. — Explanation: Soil loss increased, water became cloudy, and fish numbers decreased after tree removal.

GED-SCI1-021 — Correct: B — Correct Answer: Brown insects were more likely to survive and reproduce in that environment. — Explanation: Natural selection favors traits that improve survival and reproduction in a specific environment.

GED-SCI1-022 — Correct: C — Correct Answer: Distance is directly proportional to time. — Explanation: The distance increases by 343 m for each additional second, showing a constant proportional relationship.

GED-SCI1-023 — Correct: 50 — Tolerance: exact — Correct Answer: 50 — Explanation: The increase is 6 liters. Percent increase = $6 \div 12 \times 100 = 50\%$.

GED-SCI1-024 — Correct: C — Correct Answer: Patient C — Explanation: Patient C has the highest listed resting heart rate at 81 beats/min.

GED-SCI1-025 — Correct: C — Correct Answer: Friction — Explanation: Wheels reduce friction between the box and the floor, making the box easier to move.

GED-SCI1-026 — Correct: B — Correct Answer: Amount of light — Explanation: A controlled factor is kept the same in all groups. The amount of light is the same for every tank.

GED-SCI1-027 — Correct: D1=water vapor; D2=rain or snow — Correct Answer: D1=water vapor; D2=rain or snow — Explanation: Evaporation changes liquid water into water vapor, and precipitation returns water to Earth as rain, snow, sleet, or hail.

GED-SCI1-028 — Correct: A — Correct Answer: Adding bulbs in series increases total resistance in the circuit. — Explanation: In a series circuit, adding bulbs increases total resistance, reducing current and bulb brightness.

GED-SCI1-029 — Correct: C — Correct Answer: 35°C — Explanation: The greatest number of colonies, 46, occurred at 35°C .

GED-SCI1-030 — Correct: A — Correct Answer: The crushed chalk reacts completely in less time than the large cube. — Explanation: Crushed chalk has more surface area exposed to the acid, which should increase the reaction rate.

GED-SCI1-031 — Correct: A,B — Correct Answer: Test several students instead of only one student. | Use the same exercise time for every trial. — Explanation: More participants and consistent exercise time make the investigation more reliable and better controlled.

GED-SCI1-032 — Correct: D — Correct Answer: Mars — Explanation: Mars has the greatest listed average distance from the Sun at 228 million km.

GED-SCI1-033 — Correct: T1=C; T2=B; T3=A — Correct Answer: T1=Exchanges O₂ and CO₂ with the environment; T2=Moves oxygen and nutrients through the blood; T3=Breaks down food and absorbs nutrients — Explanation: The respiratory system exchanges gases, the circulatory system transports materials in blood, and the digestive system breaks down food and absorbs nutrients.

GED-SCI1-034 — Correct: B — Correct Answer: 5°C/min — Explanation: Temperature increased from 21°C to 51°C, a 30°C increase over 6 minutes. The average rate is $30 \div 6 = 5^\circ\text{C}/\text{min}$.